

Chapter - 1

Introduction of Java language

What is Code?

→ Our Computers understand only Binary language, because they are electric systems.

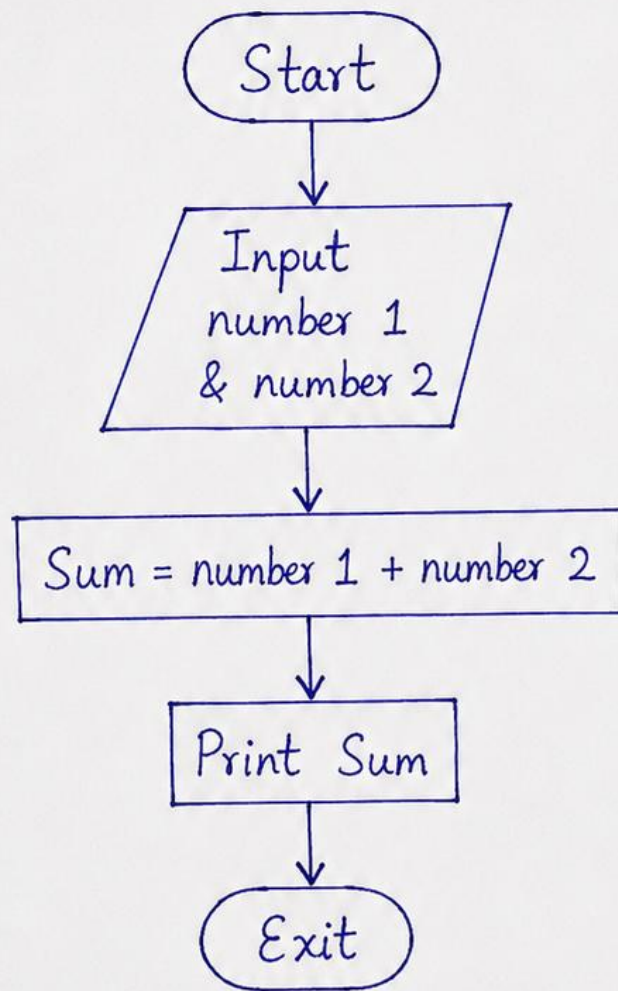
→ If current flows then it's 1 and if no current flows it's 0

Example

Decimal	→	Binary
0	→	0
1	→	1
3	→	11
4	→	100

Pseudocode and Flowchart

- ① Start
- ② Input 2 numbers
- ③ Calculate $\text{Sum} = \text{number 1} + \text{number 2}$
- ④ Print Sum
- ⑤ Exit



Installation

- ① Java Development kit (JDK)
- ② Code Editor / IDE (Integrated Development Env.)
 - ├ Visual Studio Code
 - ├ IntelliJ
 - └ Eclipse

First Code:

```
class First_class {  
    public static void main(String args[]) {  
        System.out.println("Hello World");  
    }  
}
```

→ Always save file with the create name of the class in this case: FirstClass.java

Question 1

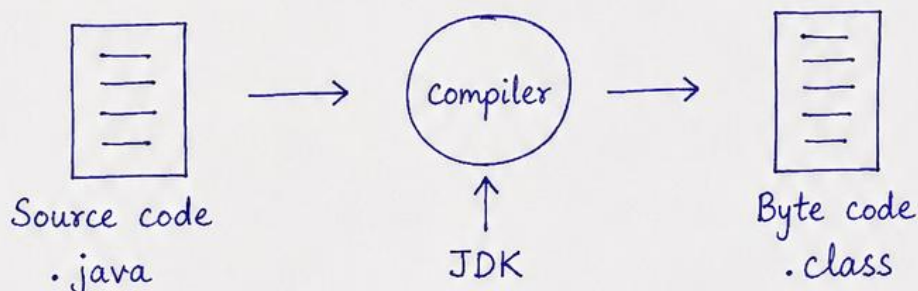
How is java code run?

→ The java code runs on two main steps

(i) Compilation (ii) Execution

JDK	JRE : Java Runtime Environment
 + Lib + Development tools	JVM = Java Virtual Machine

→ In compilation the source code that is saved as: (.java) that is given to the compiler & then it is saved in Byte form as (.class)



→ In execution the saved byte code is further given to the JVM (Java Virtual Machine) and then further given to native code.

→ Native code means the code understood by the system & the environment.



Components of Code

① Function

→ Function is a component that is used to perform the given code.

```
void main(){
    function
}
```

② Class

→ Class is a big entity which holds more than one function in a class.

```
class Main {
    void main(){
    }
}
```



Boilers Plate Code

→ This type of code is by default come in the IDE (Integrated Development Environment)

Example package com.apnacollege;

Input

```
[ public class Main {  
    public static void main(String[] args){  
        // output  
        System.out.print ("Hello World");  
        System.out.println ("Hello World");  
        System.out.print ("Hello World\n");  
    }  
}
```

output

```
[ Hello World  
  Hello World  
  Hello World
```

Question 2

→ Point the pattern

*
* *
* * *

```
package com.package;  
public class Main {  
    public static void main(String[] args) {  
        // output  
        System.out.println("*");  
        System.out.println("**");  
        System.out.println("***");  
    }  
}
```

Variables

→ Java is the language and in this variables are the building blocks.

Example

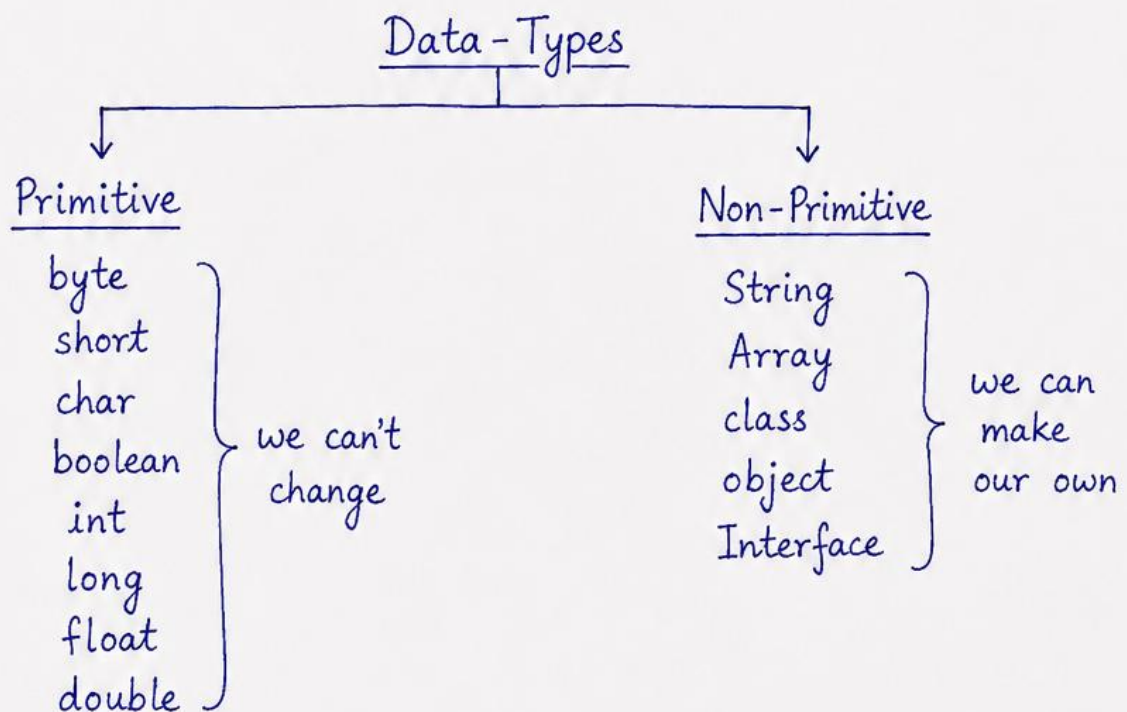
$a = 25$ $b = 10$
 $2 * (a + b)$
 ↑ ↑
constant variable

→ Basically to store data memory is the main component of a computer they are saved as small packets of data

Second Code

```
package com.package
public class Main {
    public static void main(String[] args) {
        // variables
        String name = "XYZ";
        int a = 25;
        int b = 10;
        int age = 18;
        double price = 25.25;
    }
}
```

Data Types



→ 1 byte is equal to 8 bits

Third Code

```
package com.package  
public class Main {  
    public static void main (String[] args) {  
        // variables  
        int a = 10;  
        int b = 25;  
        int sum = a + b;  
        System.out.println (sum);  
    }  
}
```

Fourth Code

```
package com.package  
public class Main {  
    public static void main (String[] args) {  
        // variables  
        int a = 10;  
        int b = 25;  
        int diff = a - b;  
        System.out.println (diff);  
    }  
}
```

→ Same with Multiplication and Division.

Input in Java

- As we know about output now we have the input to receive output hence we have a code (System.in).
- Now we have to store at local storage with code (next) and print the class.

Example

```
Scanner sc = new Scanner(System.in);  
String name = sc.next();  
System.out.println(name);
```

- Here next is for one word and nextline is for full one line or word.

Fifth Code

```
package com.package  
import java.util.*;  
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int a = sc.nextInt();  
        int b = sc.nextInt();  
        int sum = a + b;  
        System.out.println(sum);  
    }  
}
```

Conditional Statement

→ This is a statement that has valid reasons and lines. Here are different reasons we can use like :-

① if else ② else if ③ switch ④ break

① If, else syntax.

```
if (condition) {  
    stt 3;  
    stt 4;  
} else {  
    stt 5;  
    stt 6;  
}
```

Sixth Code

```
import java.util.*;  
public class Conditions {  
    public static void main (String[] args) {  
        Scanners sc = new Scanner (System.in);  
        int age = sc.nextInt();  
        if (age ≥ 18) {  
            System.out.println ("Adult");  
        } else {  
            System.out.println ("Not Adult");  
        }  
    }  
}
```

Seventh Code

```
import java.util.*;
public class Conditions {
    public static void main (String args[]) {
        Scanner sc = new Scanner (System.in);
        int x = sc.nextInt();
        if (x % 2 == 0) {
            System.out.println ("Even");
        } else {
            System.out.println ("Odd");
        }
    }
}
```

Eighth Code

```
import java.util.*;
public class Conditions {
    public static void main (String args[]) {
        Scanner sc = new Scanner (System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();
        if (a == b) {
            System.out.println ("Equal");
        } else if (a > b) {
            System.out.println ("a is greater");
        } else {
            System.out.println ("a is smaller");
        }
    }
}
```

Ninth Code

```
import java.util.*;
public class Conditions {
    public static void main (String args[]) {
        Scanner sc = new Scanner (System.in);
        int a = sc.nextInt();
        int b = sc.nextInt();
        if (a >= b) {
            System.out.println ("a is greater");
        } else {
            System.out.println ("a is smaller");
        }
    }
}
```

② else if syntax.

```
if (condition 1) {  
    stt 1;  
    stt 2;  
} else if (condition 2) {  
    stt 3;  
    stt 4;  
} else {  
    stt 5;  
    stt 6;  
}
```

Ninth Code

```
import java.util.*;  
public class Conditions {  
    public static void main (String args[]) {  
        Scanner sc = new Scanner (System.in);  
        int a = sc.nextInt();  
        int b = sc.nextInt();  
        if (a == b) {  
            System.out.println ("Equal");  
        } else if (a > b) {  
            System.out.println ("a is greater");  
        } else {  
            System.out.println ("a is smaller");  
        }  
    }  
}
```

Switch

→ Switch is like a button in which the input given is managed by the cases and from the result and output is given.

Syntax of Switch,

```
switch (variable) {  
    case 1 :  
        stt 1;  
        stt 2;  
        break;  
    case 2 :  
        stt 3;  
        break;  
    default :  
        stt 4;  
}
```

→ We can add any variable instead of any alphabet also, like given syntax is by default we can make changes.

Tenth Code

```
import java.util.*;
public class Conditions {
    public static void main (String args[]) {
        Scanner sc = new Scanner (System.in);
        int button = sc.nextInt();
        switch (button) {
            case 1: System.out.println ("Hello");
                    break ;
            case 2: System.out.println ("Namaste");
                    break ;
            case 3: System.out.println ("Bonjour");
                    break ;
            default : System.out.println ("Invalid");
        }
    }
}
```

Loop

- Loop is the system of the java in which we can repeat the function constantly.
- There are three loop options:
 - ① for loop ② while loop ③ do while loop

Syntax for Loop

```
for (condition) {  
    // do same thing  
}
```

- Here the condition is the condition variable initially it is zero, after printing the count increases.
- Condition is the time for the function i.e for what time the function should run
- All printing commands are written between the brackets for adding multiple functions

Eleventh Code

```
public class Loops {  
    public static void main (String args[]) {  
        for (int counter = 0 ; counter <= 3 ;  
            counter = counter + 1);  
        System.out.println ("Hello World");  
    }  
}
```

Question-3

Print numbers 0 to 10

Ans

```
public class Loops {  
    public static void main (String args[]) {  
        for (int i = 0 ; i < 11 ; i++) {  
            System.out.println (i);  
        }  
    }  
}
```

Syntax of while loop

```
while (condition) {  
    // do something  
}
```

→ While loop is same as for loop in this the function will not stop till the defined value is reached.

Twelveth Code

```
public class Loops {  
    public static void main (String args[]) {  
        int i = 0;  
        while (i < 11) {  
            System.out.println(i);  
            i = i + 1; // i++  
        }  
    }  
}
```

→ This same thing can also be done by do-while loop the syntax is,

```
do {  
    // do something  
} while (condition);
```

Question 4

Print the sum of first n natural numbers

Ans

```
public class Loops {  
    public static void main (String args[]) {  
        Scanner sc = new Scanner (System.in);  
        int n = sc.nextInt();  
        int sum = 0;  
        for (int i = 0; i <= n; i++) {  
            sum = sum + i;  
        }  
        System.out.println (sum);  
    }  
}
```

Pattern Questions

1) Print the pattern

```
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

Ans

```
import java.util.*;  
public class Patterns {  
    public static void main (String args[]) {  
        int n = 4;  
        int m = 5;  
        for (int i = 1; i <= n; i++) {  
            for (int j = 1; j <= m; j++) {  
                System.out.print ("*");  
            }  
            System.out.println ();  
        }  
    }  
}
```

2) Print the pattern

```
* * * * *
*           *
*           *
* * * * *
```

Ans

```
import java.util.*;
public class Patterns {
    public static void main (String args[]) {
        int n = 4;
        int m = 5;
        // outer loop
        for (int i = 1; i <= n; i++) {
            // inner loop
            for (int j = 1; j <= m; j++) {
                // condition
                if (i == 1 || i == n || j == 1 || j == m) {
                    System.out.print("*");
                } else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

3) Print the Pattern

```
*  
* *  
* * *  
* * * *
```

Ans

```
import java.util.*;  
  
public class Patterns {  
    public static void main (String args[]) {  
        int n = 4;  
        // outer loop  
        for (int i = 1; i <= n; i++) {  
            // inner loop  
            for (int j = 1; j <= i; j++) {  
                System.out.print ("*");  
            }  
            System.out.println();  
        }  
    }  
}
```

4) Print the Pattern

```
  *
 * *
* * *
* * * *
```

Ans

```
import java.util.*;
```

```
public class Patterns {
```

```
    public static void main (String args[]) {
```

```
        int n = 4;
```

```
        // outer loop
```

```
        for (int i = 1; i <= n; i++) {
```

```
            // inner loop
```

```
            for (int j = 1; j <= i; j++) {
```

```
                System.out.print ("*");
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }
```

```
}
```

5) Print the Pattern

```
*
* *
* * *
* * * *
* * * * *
```

Ans

```
import java.util.*;
public class Patterns {
    public static void main (String args[]) {
        int n = 4;
        // outer loop
        for (int i = 1; i <= n; i++) {
            // inner loop
            for (int j = 1; j <= i; j++) {
                System.out.print ("*");
            }
            System.out.println();
        }
    }
}
```

6) Print Pattern

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Ans

```
import java.util.*;

public class Patterns {
    public static void main (String args[]) {
        int n = 5;
        // outer loop
        for (int i = 1; i <= n; i++) {
            // inner loop
            for (int j = 1; j <= i; j++) {
                System.out.print (j + " ");
            }
            System.out.println ();
        }
    }
}
```

7) Print the Pattern

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Ans

```
import java.util.*;
public class Patterns {
    public static void main(String args[]) {
        int n = 5;
        // outer loop
        for (int i = 1; i <= n; i++) {
            // inner loop
            for (int j = 1; j <= n - i + 1; j++) {
                System.out.print(j);
            }
            System.out.println();
        }
    }
}
```

8) Print the Pattern.

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Ans

```
import java.util.*
public class Patterns {
    public static void main(String args[]) {
        int n = 5;
        int number = 1;
        // outer loop
        for (int i = 1; i <= n; i++) {
            // inner loop
            for (int j = 1; j <= i; j++) {
                System.out.print(number);
                number++;
            }
            System.out.println();
        }
    }
}
```

9) Print the Pattern

```

1
0 1
1 0 1
0 1 0 1
1 0 1 0 1

```

Ans

```

import java.util.*
public class Patterns {
    public static void main (String args[]) {
        int n=5;
        // outer loop
        for (int i = 1; i <= n; i++) {
            // inner loop
            for (int j = 1; j <= i; j++) {
                int sum = i + j;
                if (sum % 2 == 0) { // even
                    System.out.print ("1");
                } else { // odd
                    System.out.print ("0");
                }
            }
            System.out.println();
        }
    }
}

```

Functions

→ Functions are the blocks of code those who take input some do operation and give out like thermometer.

Syntax

```
returnType functionName (type args1, type args 2....){  
    // operations  
}
```

→ Here returnType is DataType (int, float, string). The functionName cannot be java's keyword. Args is like input we have to define the arguments then. names.

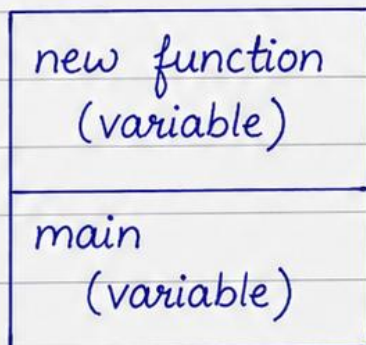
Example

Print a given name in a function

```
import java.util.*  
public class Functions {  
    public static void printMyName (String name){  
        System.out.println (name);  
        return;  
    }  
  
    public static void main (String args[]){  
        Scanner sc = new Scanner (System.in);  
        String name = sc.next();  
print MyName = sc.next ();  
        print MyName (name);  
    }  
}
```

What Happens In Memory

- When we create a function (main) who called a new function. Then firstly all functions are saved in a stack inside the memory.
- Inside stack the things are called stack frame. Then first stack frame will be main function. Inside all variables are stored in the main function.
- If assume we have created one more new function and then when it returns to the first main function this new function is removed.



Question

Make a function to add 2 numbers and return their sum.

Ans

```
public class Functions {  
    public static int calculateSum(int a, int b) {  
        int sum = a + b;  
        return sum;  
    }  
  
    public static void main(String args[]) {  
        Scanner sc = new Scanner(System.in);  
        int a = sc.nextInt();  
        int b = sc.nextInt();  
        int sum = calculateSum(a, b);  
        System.out.println("Sum of 2 no is: " + sum);  
    }  
}
```

Time And Space Complexity

- The use of Time and space complexity is for proper management of time so that we can make code run
- The Relation between Input Size and Running Time Operation.
- The larger the input size the longer the code.

Example

```
① public static void main(String args[]) {  
    Scanner sc = new Scanner(System.in);  
    int n = sc.nextInt();  
    for (int i = 0; i < n; i++) {  
        System.out.println("Hello");  
    }  
}
```

time complexity $\propto$ input n
↓
linear Relation

```
② public static void main (String args[]) {  
    Scanner sc = new Scanner (System.in);  
    int n = sc.nextInt();  
    for (int i=0; i<n; i++) {  
        for (int j=0; j<n; j++) {  
            System.out.println ("Hello");  
        }  
    }  
}
```

Arrays

→ When we have to store many or more variables of same DataType then we store them in a list and then we can easily use in code.

Syntax

```
① type [] arrayName = new type [size];  
                        ↑  
                    keyword
```

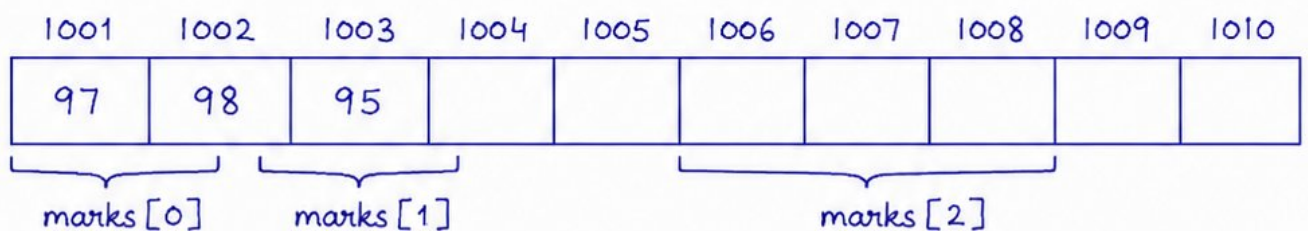
```
② type [] arrayName = {1,2,3,4,5,6};
```

Two tenth Code

```
import java.util.*;
```

```
public class Arrayy {  
    public static void main (String args[]) {  
        int [] marks = new int [3];  
        marks[0] = 97; // phy  
        marks[1] = 98; // chem  
        marks[2] = 95; // eng  
        for (int i = 0 ; i < 3 ; i++) {  
            System.out.println (marks[i]);  
        }  
    }  
}
```

② What happens In Memory



```
int marks[] = new int [3];
```